SEQUENCES, SERIES AND MEANS ON ONE PAGE

Add a step, or multiply by one

Taking logarithms turns every geometric pattern on this page into an arithmetic one.

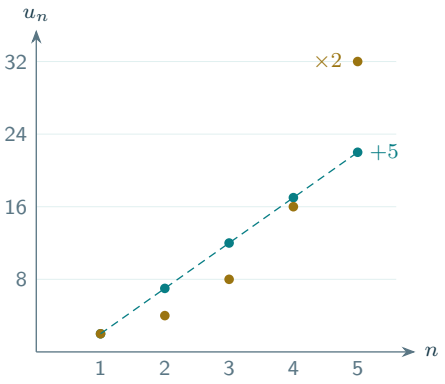
THE TWO FAMILIES

A common difference d, or a common ratio r

	arithmetic	geometric
nth term	$u_n = u_1 + (n-1)d$	$u_n = u_1 r^{n-1}$
sum of n	$S_n = \frac{n}{2}(u_1 + u_n)$	$S_n = \dfrac{u_1(1-r^n)}{1-r}$
to infinity	diverges, unless every term is 0	$S_\infty = \dfrac{u_1}{1-r}$ when $\lvert r\rvert < 1$

$$u_n = u_1 r^{n-1} \implies \ln u_n = \ln u_1 + (n-1)\ln r$$

the logs of a positive geometric sequence form an arithmetic sequence with $d = \ln r$



$2, 7, 12, 17, 22$ and $2, 4, 8, 16, 32$: equal steps up, or equal ratios.

ARITHMETIC SUM

Pair the ends

Write the sum forwards and backwards; each column adds to the same total $u_1 + u_n$, and there are n columns, so

$$2S_n = n(u_1 + u_n) \implies S_n = \tfrac{n}{2}(u_1 + u_n),$$

which for $1 + 2 + \cdots + 100$ gives $50 \times 101 = 5050$.

GEOMETRIC SUM

Shift and subtract

Multiplying by r slides every term one place along, so only two terms survive the subtraction:

$$S_n - rS_n = u_1 - u_1 r^n \implies S_n = \frac{u_1(1-r^n)}{1-r},$$

and when $\lvert r\rvert < 1$ the term r^n dies away, leaving $S_\infty = \dfrac{u_1}{1-r}$.

THE MEANS

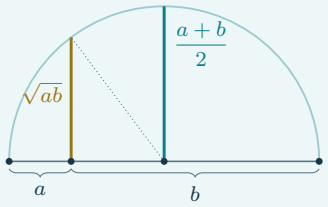
The middle term of each family

Solving $m - a = b - m$ and $m/a = b/m$ for the middle term gives the two means of positive a and b:

$$\text{AM} = \frac{a+b}{2}, \qquad \text{GM} = \sqrt{ab}.$$

Between 2 and 18 these give $2, 10, 18$ and $2, 6, 18$. The geometric mean is the arithmetic mean of the logs, since $\ln\sqrt{ab} = \frac{1}{2}(\ln a + \ln b)$, and expanding $\left(\sqrt{a} - \sqrt{b}\right)^2 \geq 0$ gives

$$\frac{a+b}{2} \geq \sqrt{ab}, \quad \text{with equality only when } a = b.$$



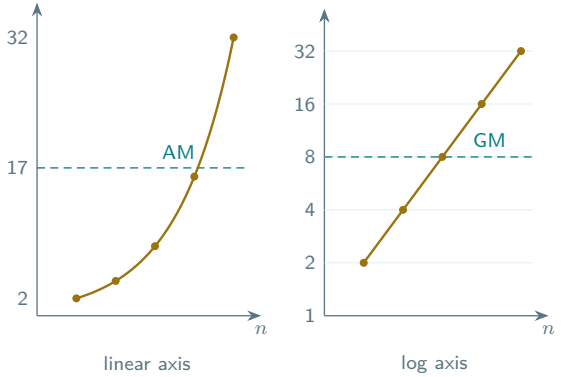
The radius is the AM and the half-chord is the GM, so the half-chord is never longer.

LINEAR AND SEMI-LOG GRAPHS

Each family has an axis that straightens it

On linear axes an arithmetic sequence is a straight line of gradient d and a geometric one curves upward; a semi-log graph spaces the vertical axis by $\log u$, so equal ratios sit at equal heights and the geometric sequence straightens to gradient $\log r$, which is the bridge above drawn as a picture.

Halfway up the linear axis between 2 and 32 is the AM, 17, and halfway up the log axis is the GM, 8, which is also the term at the middle step $n = 3$.



The same data, $u_n = 2^n$, on both axes.

Conventions. Here u_1 is the first term (some texts write a) and S_n is the sum of the first n terms. A semi-log graph has one logarithmic axis and a log-log graph has two; plotting $\lg u$ or $\ln u$ on an ordinary axis gives the same straight line, with gradient $\lg r$ or $\ln r$ respectively.